

Digital System and Design using Verilog



MINI PROJECT

Topic: LCD interfacing with FPGA
to display rolling message

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Verilog code:

```
module led(  
    input wire clk,  
    input wire reset,  
    output reg [7:0] lcd_data,  
    output reg lcd_rs,  
    output reg lcd_en  
);  
    reg [6:0] message [0:15]; // Message array  
    reg [3:0] pos; // Current position in message  
    integer i;  
  
    initial begin  
        message[0] = "H"; message[1] = "e"; message[2] = "l"; message[3] =  
"l";  
        message[4] = "o"; message[5] = " "; message[6] = "W"; message[7] =  
"o";  
        message[8] = "r"; message[9] = "l"; message[10] = "d"; message[11]  
= "!";  
        message[12] = " "; message[13] = " "; message[14] = " ";  
message[15] = " ";  
        pos = 0;  
    end  
  
    always @(posedge clk or posedge reset) begin  
        if (reset) begin  
            pos <= 0;  
        end else begin  
            lcd_data <= {message[pos], 8'b00000000}; // Send character to  
LCD  
            lcd_rs <= 1; // Data mode  
            lcd_en <= 1; // Enable signal  
  
            if (pos < 15) pos <= pos + 1; // Scroll through message  
            else pos <= 0; // Loop back to start  
        end  
    end  
endmodule
```

Testbench:

```
module tb_led;

    // Declare signals for the lcd_scroll module
    reg clk;
    reg reset;
    wire [7:0] lcd_data;
    wire lcd_rs;
    wire lcd_en;

    // Instantiate the lcd_scroll module
    led uut (
        .clk(clk),
        .reset(reset),
        .lcd_data(lcd_data),
        .lcd_rs(lcd_rs),
        .lcd_en(lcd_en)
    );

    // Clock generation
    always #5 clk = ~clk; // 100MHz clock (10ns period)

    initial begin
        // Initialize signals
        clk = 0;
        reset = 1;

        // Apply reset for some time
        #10 reset = 0; // Release reset after 10ns

        // Monitor output signals
        $monitor("Time = %0d, lcd_data = %0h, lcd_rs = %b, lcd_en = %b",
$time, lcd_data, lcd_rs, lcd_en);

        // Simulate for a sufficient period to see the scrolling
        #200;

        // End the simulation

    end
endmodule
```

Output:

